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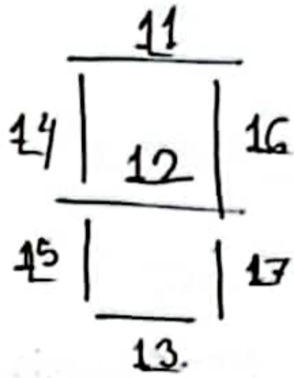
CSE-4205:

DLD:

7 segment display / digit representator.

Truth Table:



Number	L1	L2	L3	L4	L5	L6	L7
0	1	0	1	1	1	1	1
1	0	0	0	0	0	~~0~~1	1
2	1	1	1	0	1	1	0
3	1	1	1	0	0	1	1
4	0	1	0	1	0	1	1.
5	1	1	1	1	0	0	1
6.	1	1	1	1	1	0	1
7	1	0	0	0	0	1	1
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1.

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Algebraic Expression:

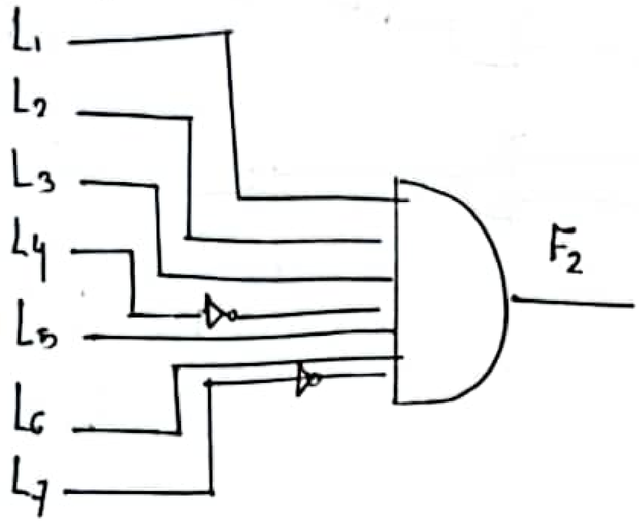
$$F_2 = L_1 . L_2 . L_3 . L_4' . L_5 . L_6 . L_7'$$

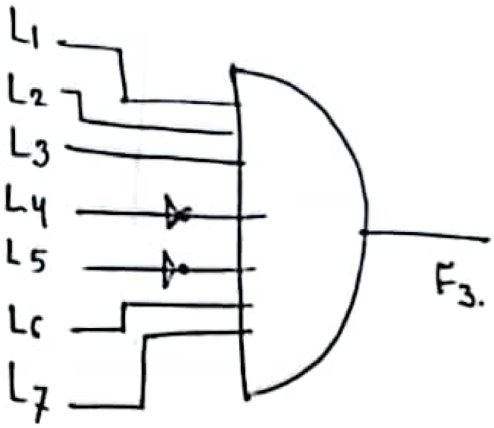
function

prime for 0.

$$F_1 = L_1' . L_2' . L_3' . L_4' . L_5' L_6 . L_7$$

Logic Gates:





a+b:

(Augend) a	(Addend) b	(Sum) s	(Carry) c
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$S = a'b + ab'$

$c = ab.$

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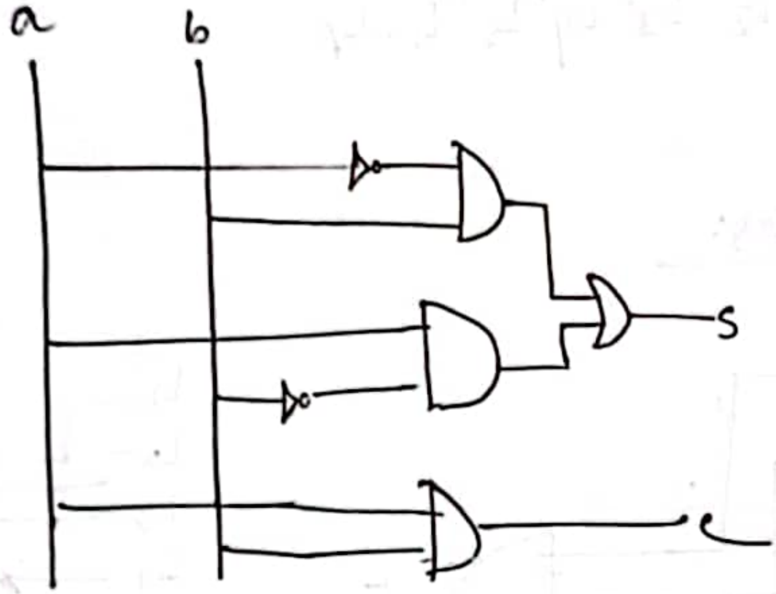
* Perform Add Operation:

2 inputs as Augend and Addend
2 outputs as Sum and Carry

**



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CSE-4205

Digital System:

1. Preserving data of 0s and 1s.
2. Electrical signals ⟹ (Continuous)
3. Math Operation.
4. Discrete Signal

⟹ Embedded Digital Computer.

- MIPS ⟶ Million of Instructions Per Second

Conversion:

$\left(\sum x_i \times b^i\right)_b$ 407.322

mantissa ⟶ integer values (407322)
exp. ⟶ (−3)

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CSE-4205:

$$\begin{array}{r l} 243 & \longrightarrow \text{Minuend} \\ -149 & \longrightarrow \text{Subtrahand} \\ \hline 94 & \end{array}$$

*** Complement:**

$A = A'$

such that $1 = A + A'$

→ To perform distribution.

(RADIX COMPLEMENT) $\boxed{r\text{'s complement}}$

$\boxed{(r-1)\text{'s complement}}$ (DIMINISHED RADIX COMPLEMENT)

$r \rightarrow$ base / radix

A-B.
↳ A + (-B)

*** (r-1)'s complement:**

$$\boxed{(r-1)\text{'s complement of } N = r^n - r^{-m} - N}$$

where,
$n \rightarrow$ no. of digits in integer
$m \rightarrow$ no. of digits in fraction

Ex: → 1. 9's complement of $(52520)_{10}$

$= 10^5 - 10^0 - 52520$

$= 47479$

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2. 9's comp. of $25.639 = 10^2 - 10^{-3} - 25.639$

$= 74.360$

3. 1's complement of $11011.01 = 2^5 - 2^{-2} - (11011.01)$ make it decimal

$= 00100.10$

4. 1's complement of $11.101 = 2^2 - 2^{-3} - (11.101)$ make it decimal

$= 00.010$

* Subtract each digit from $(r-1)$ → SHORTCUT.

$$\begin{array}{r} 99.999 \\ -\ 25.639 \\ \hline 74.360 \end{array}$$

* r's complement:

$\boxed{r\text{'s complement of } N = r^n - N}$

SHORTCUT ⟶ 1. leave the LSB zero from right
2. subtract second digit from r.
3. subtract rest digits from r-1.

$52520 \rightarrow \begin{array}{ccccc} 9 & 9 & 9 & 10 & 0 \\ 5 & 2 & 5 & 2 & 0 \end{array} \longrightarrow 47480$

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* Subtraction with r's complement :

$$M - N = M + N'$$

→ discard the carry

Q. $(72532)_{10}$ → 72532
$(03250)_{10}$ → + 96750

(1)69282

↓

discard the carry

Q. $(1010100)_2$ → 1010100
$(1000100)_2$ → 0111020

Q. 03250
− 72532

27468

−30718

1000100
− 1010100

0101100

1110000

− 0010000

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* Subtraction with (r-1)'s complement:

$$
\begin{array}{r}
72532 \\
-03250 \\
\hline
+96749 \\
\hline
169281 \\
+1 \\
\hline
69282
\end{array}
$$

Ans.

2. 30717 → -(69282)

3.

CamScanner

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CSE-4205:

* Overflow → extra bit
→ erroneous result

(10) - 1010
(11) = 1011

$\boxed{1}$0101

→ Unsigned Overflow
↳ Result MSB same.

float addend
float augend
float result

⇒ Correction: → use a register.

			S	C
0 0	→		0	0
0 1	→		1	0
1 0	→		1	0
1 1	→		0	1

+2 → 0010
+3 → 0011

0101.

MSB
↳ 0
positive number

* Signed Magnitude System:

−3 = 1011
−2 = 1010

$\boxed{1}$ 0101.
↳ discarded.

Value becomes +5

→ Signed Overflow
↳ Result

* Signed Complement System

−3 → 1101
−2 → 1110

$\boxed{1}$1011
↓ 0101.

* Underflow

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* Arithmetic Addition:

±6

±13

00000110
11111001
+1
0
00001101
11110010
1 1

① 6 = 0110
13 = 1101
10011

② 6 = 0110
-13 = 0011.
1001

③ -6 = 11111010
13 = 00001101
100000111

④ -6 = 11111010
-13 = 11110011
100001101

Ans.

① 6 = 00000110
13 = 00001101
00010011

② 6 = 00000110
-13 = 11110011
11111001

Ans.

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Binary Code:

23 → 10111. → | 1 | 1 | 1 | 0 | 1 |

Binary Coded Decimal:

0 → 0000
1 → 0001
2 → 0010
3 → 0011
4 → 0100
5 → 0101
6 → 0110
7 → 0111
8 → 1000
9 → 1001.

7 = 0111
8 = 1000

1111 > 9 → invalid in BCD.

(add 6) 0110

0001 0101

(so. increase the number of bits)

2 = 0010
3 = 0011.

0101 ≤ 9

Weighted Code:

184 = 0001 1000 0100
+576 = 0101 0111 0110

0110 1111 1010
+ 0110

700 0111 0000 0000

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184	=	0001	1000	0100
+ 576	=	0101	0111	0110
		0111	10000	1010
			0110	0110
$\boxed{760}$	=	0111	0110	0000 =

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CSE-4205:

⟹ Weighted Code:

if 1 → consider
if 0 → don't consider.

2^3 2^2 2^1 2^0
↓ ↓ ↓ ↓
0 1 0 1 $= 2^2+2^0 = 4+1 = 5.$

2421

0000 = 0
0001 = 1
0010 = 2
1000

maintain ← (??) to get 9's complement.

8 4 -2 -1

0000 = 0
0111 = 1
0110 = 2

BCD+3

0 = 0011
1 = 0100
2 = 0101
3 = 0110

50 43 210

0100010 = 0
0100010 = 1
0100100 = 2
0101000 = 3
0110000 = 4

1
1
1
1
1

Repeated

5-9

* Biquinary

→ Used to create 9's complement easily (excess-3, 84-2-1, 2421) ⇒ Self complementary Code

$\boxed{0011 \rightarrow 9 \rightarrow 1100}$ → In excess 3 method.

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* Gray Code:

$3 = 011 \rightarrow t_i \cdot R$ $\boxed{0|1|1}$

$4 = 100 \rightarrow t_{i+1}$ $t1$ $\boxed{1|1|1}$

$t2$ $\boxed{1|0|1}$

$t3$ $\boxed{1|0|0}$

$\boxed{t_{i+1} + t_g \rightarrow \text{delay}}$

$\rightarrow$ decrease the delay

For example:

$t_{i+2} = R_2 = R_1 \times 2$

* Alphanumeric Code:

$\rightarrow$ ASCII

* Binary $\longleftrightarrow$ Grey:

$(11011)_{Grey} = Bin.$

$(101011)_{Bin} = Grey.$

1 0 1 0 1 1

1 1 1 1 1 0 $\rightarrow$ Gray.

1 1 0 1 1

$\downarrow$

~~1 0 1 1 0~~

* Error Detection Code:

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CSE-4205:

* Set of elements $\rightarrow \{0, \ldots, 9\}$
* Set of operations.
* Axioms / Postulates $\longrightarrow$ Assumption.

POSTULATES:

1. Closure
2. Associative law
3. Commutative law
4. Identity Element.
5. Inverse
6. Distributive law

Boolean Algebra:

1. Closure
2. Identity element.
3. Commutative
4. Distributed
5. Complement.
6. Two elements not equal to each other.

* Duality:

$\therefore x+(y+z) = (x+y)+z$

x	y	z	$x+y$	$y+z$	$x+(y+z)$	$(x+y)+z$
1	1	1	1	1	1	1
1	1	0	1	1	1	1
1	0	1	1	1	1	1
1	0	0	1	0	1	1
0	1	1	1	1	1	1
0	1	0	1	1	1	1
0	0	1	0	1	1	1
0	0	0	0	0	0	0

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x	y	z	xy	yz	$x(yz)$	$(xy)z$
1	1	1	1	1	1	1
1	1	0	1	0	0	0
1	0	1	0	0	0	0
1	0	0	0	0	0	0
0	1	1	0	1	0	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

$\therefore x(yz) = (xy)z$

* Boolean Function:

$F_1 = x + y'z$

x	y	z	y'	$y'z$	$x+y'z$
1	1	1	0	0	1
1	1	0	0	0	1
1	0	1	1	1	1
1	0	0	1	0	1
0	1	1	0	0	0
0	1	0	0	0	0
0	0	1	1	1	1
0	0	0	1	0	0

* Algebraic Manipulation:

$$F = x(x'+y)$$

$$y = x + x'y$$

$$\Rightarrow y = (x+x')(x+y)$$

$$= 1(x+y)$$

$$= x+y$$

$$z = (x+y)(x+y')$$

$$= x + xy + xy' + yy'$$

$$= x(1+y+y')$$

$$= x$$

CSE-4203:

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$y'z = y'z.1 = y'z(x+x') = xy'z + x'y'z$

$x = x.1.1$

$= x.(y+y')(z+z')$

$= (xy+xy')(z+z')$

$= xyz + xyz' + xy'z + xy'z'$

(m_j) Minterms	x	y	z	F	
$x'y'z'$	0	0	0	0	$\to 0$
$x'y'z$	0	0	1	1	$\to 1$
$x'yz'$	0	1	0	0	$\to 2$
$x'yz$	0	1	1	0	$\to 3$
$xy'z'$	1	0	0	1	$\to 4$
$xy'z$	1	0	1	1	$\to 5$
xyz'	1	1	0	1	$\to 6$
xyz	1	1	1	1	$\to 7$

$\therefore F(x,y,z) = x+y'z$ (not canonical) (standard form)

$= xy'z + x'y'z + xyz + xy'z' + xyz'$ (canonical)

$= m_1 + m_4 + m_5 + m_6 + m_7$ Ans.

Sum of minterms $= \Sigma(1,4,5,6,7)$.

$\Sigma \to$ Summation.

~~SOP~~

$\to$ SOP: Sum of Product.

$\therefore F' = m_0 + m_2 + m_3$

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Maxterm: (M_j)

x	y	z	F		
0	0	0	0	$\longrightarrow$	$x+y+z = M_0$
0	0	1	1	$\longrightarrow$	$x+y+z' = M_1$
0	1	0	0	$\longrightarrow$	$x+y'+z = M_2$
0	1	1	0	$\longrightarrow$	$x+y'+z' = M_3$
1	0	0	1	$\longrightarrow$	$x'+y+z = M_4$
1	0	1	1	$\longrightarrow$	$x'+y+z' = M_5$
1	1	0	1	$\longrightarrow$	$x'+y'+z = M_6$
1	1	1	1	$\longrightarrow$	$x'+y'+z' = M_7$

$$\boxed{M_j = m_j'}$$

$\rightarrow$ (Maxterm and Minterm are complement to each other)

$$F' = m_0+m_2+m_3$$

$$\cancel{F = M_0+M_2+M_3} \Rightarrow F = (m_0+m_2+m_3)'$$

$$\Rightarrow F = m_0'.m_2'.m_3'$$

$$\Rightarrow F = M_0 M_2 M_3$$

$$= \prod(0,2,3).$$

$\hookrightarrow$ product.

$\longrightarrow$

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$* F = xy + x'z.$

Truth Table

x	y	z	x'	xy	x'z	xy+x'z	mj	Mj
0	0	0	1	0	0	0	*	x+y+z
0	0	1	1	0	1	1	xyz'	
0	~~0~~1	0	1	0	0	0		x+y'+z
0	1	1	1	0	1	1	xy'z'	
1	0	0	0	0	0	0		x'+y+z
1	0	1	0	0	0	0		x'+y+z'
1	1	0	0	1	0	1	xyz'	
1	1	1	0	1	0	1.	xyz	

SOP: $F = xyz' + xy'z' + xyz' + xyz.$

$= m_1 + m_3 + m_6 + m_7$

$= \Sigma(1,3,6,7).$

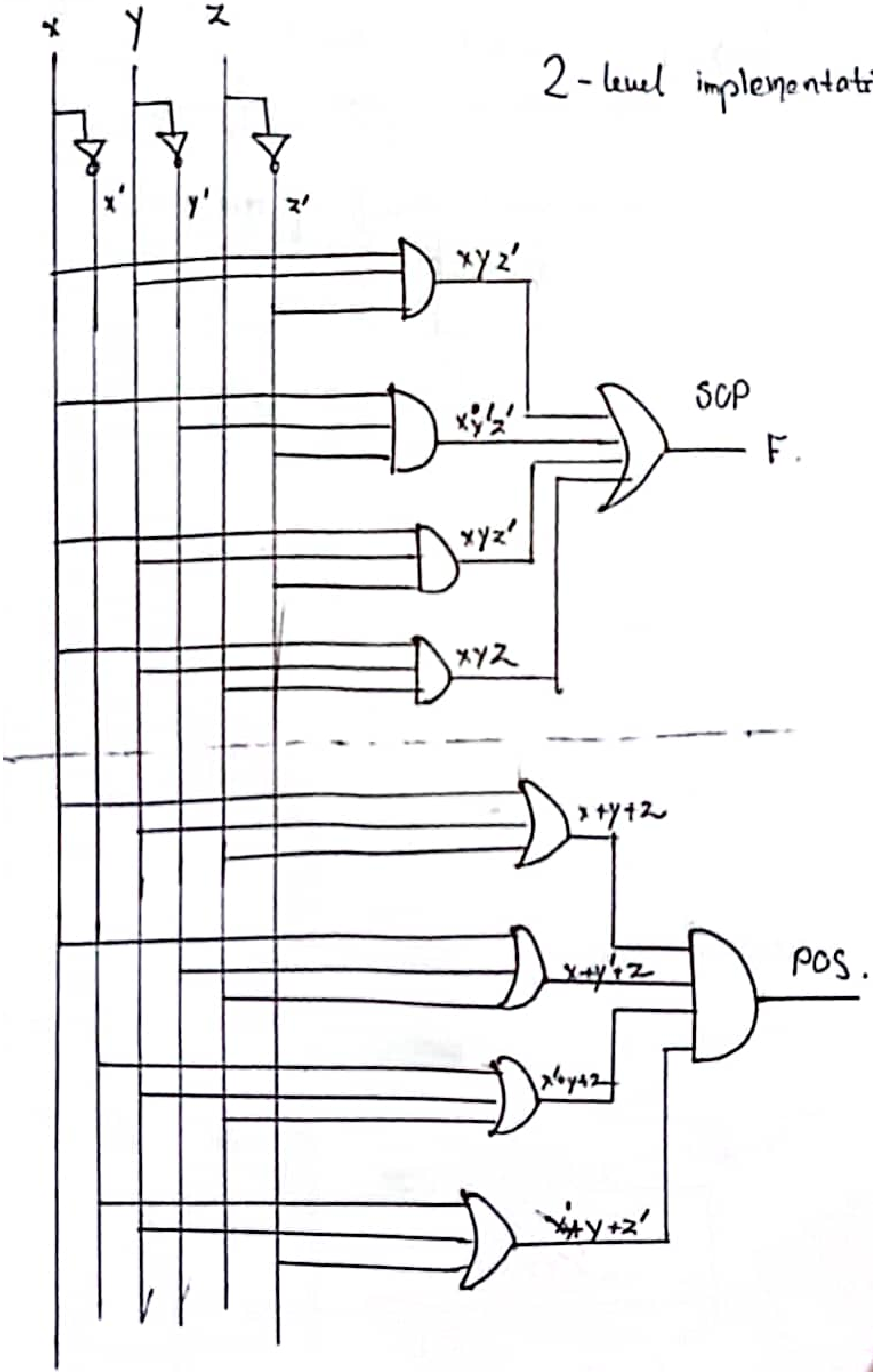
POS: $F = (x+y+z).(x+y'+z)(x'+y+z)(x'+y+z').$

$= m_0 . m_2 . m_4 . m_5.$

$= \Pi(0,2,4,5)$

Ans.

2-level implementation



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$X(xy+y'z)$

↳ non-standard format
since not in SOP/POS form.

⟶ can't be expressed in two level format.

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CSE-4205:

x	y	F
0	0	$X = 0/1 = 2$
0	1	$Y = 0/1 = 2$
1	0	$Z = 0/1 = 2$
1	1	$W = 0/1 = 2$

$2\times2\times2\times2 = 16$ functions.

→ Buffer amplifies data

Q. Is it possible to increase the no. of inputs of any logic gate?

AND $= AB$

OR $= A + B$.

NAND $= \overline{AB}$ / $A \uparrow B$.

NOR $= A \downarrow B$.

XOR $= A \oplus B$.

XNOR $= A \odot B$

→ Logic Gate
- Commutative law
- Associative law

$x+y = y+x$ | $(x+y)+z = x+(y+z)$

$xy = yx$ | $(xy)z = x(yz)$.

$\underbrace{\qquad\qquad\qquad\qquad}$

Increase the no. of inputs from 2 to ∞

→

1. NAND: Commutative Law

$$A \uparrow B = B \uparrow A.$$

2. NOR: Commutative Law

$$A \downarrow B = B \downarrow A.$$

A	B	A↑B	B↑A
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

$\therefore A \uparrow B = B \uparrow A.$

A	B	A↓B	B↓A
0	0		
0	1		
1	0		
1	1		

$\therefore A \downarrow B = B \downarrow A$

NAND $\longrightarrow$ $\overline{\text{AND}}$ $\longrightarrow$ CL/AL ✓ (Commutative/Associative Law)

NOR $\longrightarrow$ $\overline{\text{OR}}$ $\longrightarrow$ CL/AL ✓

* Parenthesis is important. *

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Chapter-3:

→ Cut and try method.
→ Drawing
→ K-map

} Simplification.

0	1
2	3

x	y	$F(x,y)$
0	0	→ $m_0 = x'y'$
0	1	→ $m_1 = x'y$
1	0	→ $m_2 = xy'$
1	1	→ $m_3 = xy$

Minterm

y →

x ↓		
0-x'	$x'y'$	$x'y$
1-x	xy'	xy
	0 y'	1 y

$F = xy$

x \ y		
	0	0
	0	1

x	y	$F = x+y$
0	0	0
0	1	1
1	0	1
1	1	1

0	1
1	1

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x	y	z	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Similar to Gray Code

00 01 10 11

$F = \Sigma(2, 3, 4, 5) \longrightarrow$ minterm.

$\longrightarrow yz$

x \ yz	00	01	11	10
0	0	001 / 1	011 / 3	010 / 2
1	100 / 4	101 / 5	111 / 7	110 / 6

m_0 0	m_1 0	m_2 1	m_3 1
m_4 1	m_5 1	m_6 0	m_7 0

$\longrightarrow m_0 = x'y'z'$

CSE-4205:

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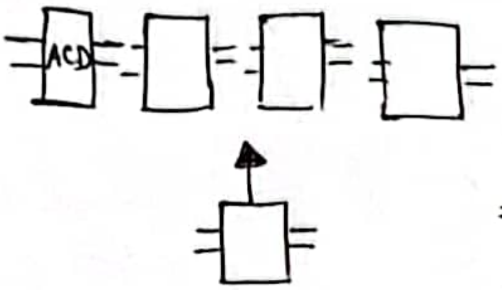
*

$Y = A'B'C + \dots\dots$

↳ simplify before implementing in hardware.

- → less input
- → less term
- → less operation.



- → Cut and try Method
- → K-map.
- → Tabular Method.

K-Map:

2 variables → 4 minterms.

X \ Y		
	xy'	$x'y$
	xy'	xy

X \ YZ	00 ($y'z'$)	01 ($y'z$)	11 (yz)	10 (yz')
x'	m_0: $x'y'z'$ 000	m_1: $x'y'z$ 001	m_2: $x'yz$ 011	m_3: $x'yz'$ 010
x	m_4: $xy'z'$ 100	m_5: $xy'z$ 101	m_7: xyz 111	m_6: xyz' 110

X \ YZ	m_0	m_1	m_2	m_3
x'	$x'y'z'$ 000	$x'y'z$ 001	$x'yz$ 011	$x'yz'$ 010
x	$xy'z'$ 100	$xy'z$ 101	xyz 111	xyz' 110
	m_4	m_5	m_7	m_6

$x \rightarrow x'/x$

$yz \rightarrow yz$

$y'z$

yz'

$y'z'$

w, x, y, z (4 variables)

wx \ yz	00	01	11	10
00	m_0 $w'x'y'z'$ 0000	m_1 $w'x'y'z$ 0001	m_3 $w'x'yz$ 0011	m_2 $w'x'yz'$ 0010
01	m_4 $w'xy'z'$ 0100	m_5 $w'xy'z$ 0101	m_7 $w'xyz$ 0111	m_6 $w'xyz'$ 0110
11	m_{12} $wxy'z'$ 1100	m_{13} $wxy'z$ 1101	m_{15} $wxyz$ 1111	m_{14} $wxyz'$ 1110
10	m_8 $wx'y'z'$ 1000	m_9 $wx'y'z$ 1001	m_{11} $wx'yz$ 1011	m_{10} $wx'yz'$ 1010

Implicant:

- adjacent squares of 2^n 1.

↳ 1, 2, 4, 8, 16 ......

$F(x,y,z) = \Sigma(2,3,4,5)$.

$P_2 = x'y$ → implicant

x \ yz	00	01	11	10
0			1	1
1	1	1		

$xy' = P_1$

implicant

4 (singles)
+2 (adjacent) → prime implicant.
6

1. Draw K-map.
2. Find out the prime implicants.

$\therefore P_1 + P_2 = xy' + x'y$

prime implicant

↳ biggest group

(i) $F(x,y,z) = \Sigma(3,4,5,6)$.

(ii) $F(x,y,z) = \Sigma(0,2,4,5,6)$.

x \ yz	00	01	11	10
0				1
1	1	1		1

$\rightarrow P_2$

$P_1 =$

· 4 singles
· 2 adjacents. → prime

6 total

(i)

x \ yz	00	01	11	10
0			1	
1	1	1		1

↳ $p = xy'$

$F = xy' + xz' + x'yz$

(ii)

x \ yz	00	01	11	10
0	1			1
1	1	1		1

$\rightarrow z'$

↳ xy'

$F = xy' + z'$.

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$F(w,x,y,z) = \sum(0,1,2,8,10,11,14,15)$

(0000, 0001, 0010, 1000, 1010, 1011, 1110, 1111)

(1) Find out the binary and catagory.

1	1		1
		1	1
1		1	1

0 - 1s → 0000 ✓
1 - 1s → 0001 ✓, 0010 ✓, 1000 ✓
2 - 1s → 1010 ✓
3 - 1s → 1011 ✓, 1110 ✓
4 - 1s. → 1111 ✓

wxyz
- × 000_
- ✓ 00_0
- ✓ _000
- ✓ _010
- ✓ 10_0
- 101_
- 1_10
- ✓ 1_11
- ✓ 111_

End Point

wxyz
- × _0_0
- × _0_0
- × 1_1_
- × 1_1_

→ Prime Implicant

$= w'x'y' + x'z' + wy$

x → unchecked